

Homework 1 - Exploring Environmental Injustices

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Load packages and read in data

```
library(tidyverse)
library(sf)
library(tmap)

# read in geodatabase of EJScreen data at the Census Block Group level
ejscreen <- sf::st_read(
  'data/ejscreen/EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb')
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
  `/Users/michaela/github/eds223/EDS223-HW1/data/ejscreen/EJSCREEN_2023_BG_StatePct_with_AS_
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:      XY
Bounding box:   xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

```
# filter to a state you are interested in
maryland <- ejscreen %>%
  dplyr::filter(ST_ABBREV == "MD")

# find the average values for all variables within counties
maryland_counties <- aggregate(maryland,
  by = list(maryland$CNTY_NAME),
  FUN = mean, na.rm = TRUE)
```

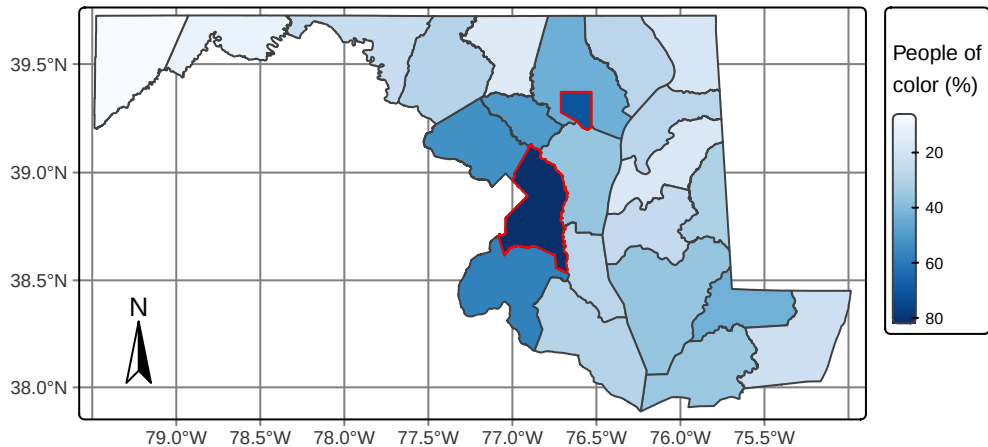
```
# filter to a county you are interested in
baltimore <- maryland_counties %>%
  dplyr::filter(Group.1 %in% c("Baltimore City"))
pg <- maryland_counties %>%
  dplyr::filter(Group.1 %in% c("Prince George's County"))
```

Visualize

Plotting the percentage of people of color in each Maryland county

```
tm_shape(maryland_counties) +
  tm_graticules() +
  tm_polygons(fill = "P_PEOPCOLORPCT", # People of color (%)
    fill.scale = tm_scale_continuous(
      values = "brewer.blues"), # create a color scale
    fill.legend = tm_legend(
      title = "\nPeople of\ncolor (%)", # break in
      # the legend title so the legend doesn't look so wide
      position = tm_pos_out(
        "right",
        "center"), # don't cover map
      title.size = 0.7,
      text.size = 0.5,
      height = 17)) +
tm_shape(baltimore, layer = "Shape") +
  tm_borders(col = "red") +
tm_shape(pg, layer = "Shape") +
  tm_borders(col = "red") +
tm_title(
  "People of Color by County, Maryland (EJScreen 2023)") +
tm_compass(
  position = tm_pos_in(
    "left",
    "bottom")) # make sure the compass doesn't cover data
```

People of Color by County, Maryland (EJScreen 2023)



This map depicts the percentage of people of color in each county in Maryland. As demonstrated by this map, Baltimore City and Prince George's County have the highest percentages of people of color. Black people make up almost 60% of the population in both of these counties (Neilsberg). Both of these counties also happen to be located in or next to two major cities, Baltimore and Washington D.C..

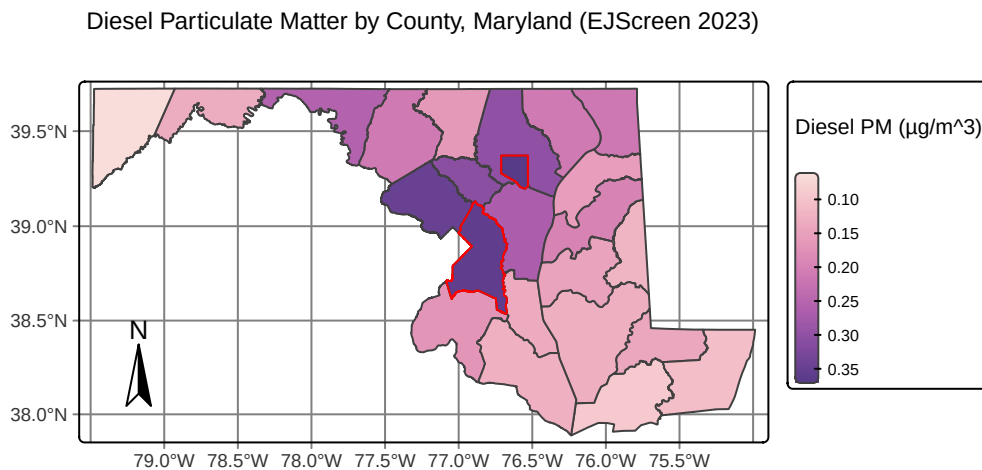
Plotting Diesel particulate matter

```
tm_shape(maryland_counties) +
  tm_graticules() +
  tm_polygons(fill = "DSLPM", # diesel pm in the air by county
    fill.scale = tm_scale_continuous(
      values = "PurpOr"), # create a color scale
    fill.legend = tm_legend(
      title = "\nDiesel PM ( $\mu\text{g}/\text{m}^3$ )\n\n",
      position = tm_pos_out(
        "right",
        "center"), # don't cover map
      title.size = 0.7,
      text.size = 0.5,
      height = 16)) +
```

```

tm_shape(baltimore, layer = "Shape") +
  tm_borders(col = "red") +
  tm_shape(pg, layer = "Shape") +
    tm_borders(col = "red") +
  tm_title("Diesel Particulate Matter by County, Maryland (EJScreen 2023)") +
  tm_compass(position = tm_pos_in(
    "left",
    "bottom")) # make sure the compass doesn't cover data

```



This map showcases the average diesel particulate matter (PM) exposure a person may encounter in each county. Diesel particulate matter is measured in $\mu\text{g}/\text{m}^3$. Diesel PM mostly comes from exhaust ([EPA](#)). We can see that diesel PM is highest in counties that also have the highest percentages of people of color.

Plotting the air toxics cancer risk

```

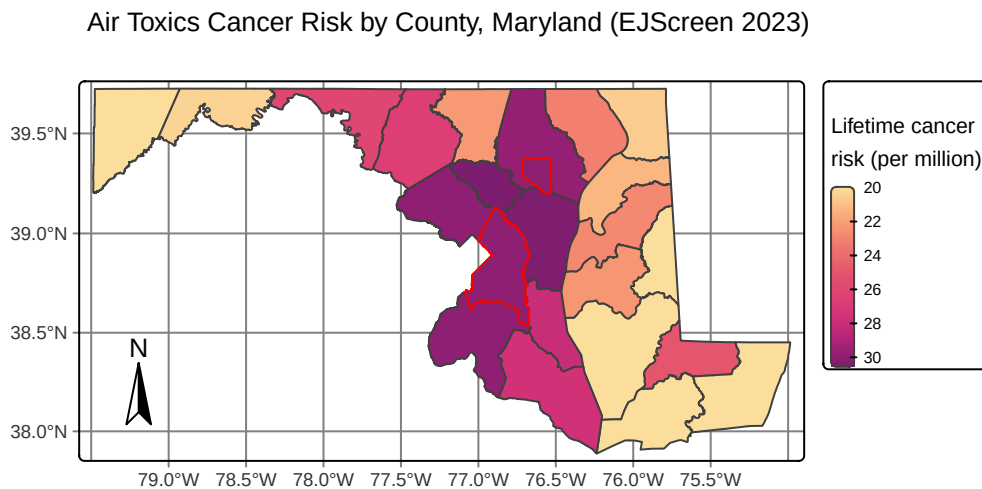
tm_shape(maryland_counties) +
  tm_graticules() +
  tm_polygons(fill = "CANCER", # air toxics inhalation cancer risk
    fill.scale = tm_scale_continuous(
      values = "SunsetDark"), # create a color scale

```

```

fill.legend = tm_legend(
  title = "\nLifetime cancer\nrisk (per million)\n",
  position = tm_pos_out(
    "right",
    "center"),
  title.size = 0.7,
  text.size = 0.5,
  height = 15)) +
tm_shape(baltimore, layer = "Shape") +
tm_borders(col = "red") +
tm_shape(pg, layer = "Shape") +
tm_borders(col = "red") +
tm_title("Air Toxics Cancer Risk by County, Maryland (EJScreen 2023)") +
tm_compass(position = tm_pos_in(
  "left",
  "bottom")) # make sure the compass doesn't cover any of the data

```



Lastly, this third map shows the air toxics cancer risk. The data shows the estimated lifetime inhalation cancer risk from carcinogens in outdoor air from the EPA's AirToxScreen ([EPA](#)). The measure of this data is persons per million over the span of 70 years (estimated one lifetime). This map serves the purpose of tying the first two maps together. People of color throughout the U.S. often face disproportionate environmental risks and thus experience higher

rates of health complications and shorter life spans. Although these data may not necessarily be causal, it is important to note the patterns.

Citations

Data

EJScreen: Environmental Justice Screening and Mapping Tool. PEDP, pedp-ejscreen.azurewebsites.net/. Accessed 4 Oct. 2026.

References

EJ Screen. “Ejscreen Map Descriptions.” US EPA, 19 Dec. 2024, web.archive.org/web/20250123161322/https://www.epa.gov/ejscreen/ejscreen-map-descriptions#ejin.

Neilsberg. Counties in Maryland Ranked by Black Population - 2025 | Neilsberg, www.neilsberg.com/insights/lists/counties-in-maryland-by-black-population/. Accessed 4 Oct. 2026.

EPA. “Air Toxics Screening Assessment.” US EPA, 7 Jan. 2025, web.archive.org/web/20250123161049/https://www.epa.gov/AirToxScreen.